RideTrace: A Modular IoT Telemetry, Safety and Emission Pre-Check Platform for Motorcycles

Project Proposal | CO3060-Sep2026 Computer Systems Design Project | **Team 14**

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Summary. We propose a low-cost, self-contained motorcycle module built on an ESP32 with an integrated 4G/GNSS modem. The *base module* tracks the rider, detects risky riding and possible crashes on-device, and alerts guardians through the cloud without needing the rider's phone. A *plug-and-play* expansion port lets optional add-ons be attached and auto-detected: a handlebar *Monitor* that displays alerts, and an *Emission Pre-Check* module that estimates CO, CO_2, HC and smoke so owners can judge whether the bike is likely to pass the official emission test before visiting a test centre. A single mobile app with role-based accounts serves both riders and guardians: a guardian simply signs in and links to a bike with a QR/invite code to receive live tracking, alerts and analytics.

1. BACKGROUND AND PROBLEM

Motorcycles are the most common personal vehicle in Sri Lanka and young riders are heavily represented in road accidents. Three gaps motivate this work:

- **Delayed response:** families often learn of a crash long after it happens; most trackers only offer location and theft alerts.
- **No behaviour insight:** speeding, aggressive acceleration and hard braking go unnoticed by guardians and riders alike.
- **Emission test failures:** owners discover a failing engine only at the test centre, paying for repeated visits. Older carburetted bikes have no OBD/CAN port to diagnose this.

2. AIM AND OBJECTIVES

Aim: design, prototype and evaluate a modular, phone-independent motorcycle safety and diagnostics platform.

O1. Measure location, speed, acceleration, lean angle and engine temperature in real time.
O2. Detect overspeed, harsh acceleration, hard braking and possible crashes *on the device*, with a rider cancel countdown.
O3. Send data and alerts over 4G (MQTT) with an SMS fallback, so the system works without the rider's phone.
O4. Provide a plug-and-play expansion bus with automatic module detection.
O5. Build a Monitor add-on (on-bike alerts) and an Emission Pre-Check add-on (CO, CO_2, HC, smoke).
O6. Build a cloud backend and a single role-based mobile app (rider/guardian) with QR/invite-code linking, live map, ride history and analytics.
O7. Quantify crash-detection accuracy and emission pre-check agreement with official test results.

3. SYSTEM ARCHITECTURE

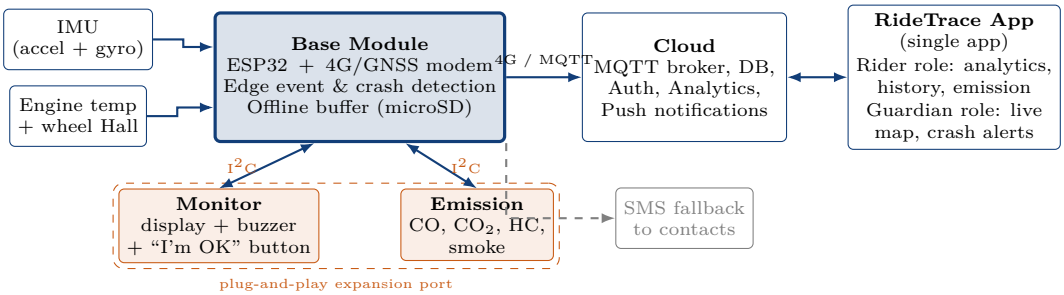


Figure 1. RideTrace architecture: base module, plug-and-play add-ons, cloud and a single role-based app.

Edge vs. cloud processing

Time-critical decisions run **on the ESP32** so they work in poor coverage: crash and risk-event detection, the cancel countdown and alert triggering. The **cloud** handles what needs storage or remote access: accounts, ride history, trends, emission analytics and guardian access. One app serves both roles: the owner registers the bike and shares a QR/invite code; anyone who signs in with it is linked as a guardian, and the owner can revoke access at any time. Data is buffered on microSD when offline and uploaded when coverage returns; crash alerts fall back to SMS.

Crash detection

A multi-condition rule: *high impact* (accel. peak) + *high angular rate* + *sudden speed drop* + *bike remains at an abnormal lean* → 30 s "Are you OK?" countdown on the Monitor/app → no response → guardians receive a push notification and SMS with the last known location. Tilt is estimated with a complementary filter; thresholds are tuned with recorded rides and controlled drop tests.

Plug-and-play expansion

Every add-on uses the same sealed connector (5 V, GND, I^2C) and carries a small ID EEPROM. At boot and periodically, the base module scans the bus, identifies the module type and version, loads the matching driver and reports it to the cloud, so the corresponding app screen appears automatically.

Emission Pre-Check module

Exhaust gas is drawn through a probe, cooling tube and water trap by a small pump into a sensor chamber. The module performs a guided idle test and reports an estimated pass/borderline/fail indication against the published limits. It is a **low-cost pre-test indicator, not a certified analyser**; it is calibrated against official test-centre readings.

4. HARDWARE COMPONENTS AND ESTIMATED COST

Unit	Components	LKR (approx.)
Base module	ESP32; SIM7600/A7670 4G modem with GNSS; IMU (MPU6050, upgradeable to BMI270); DS18B20/MAX6675 temperature; A3144 Hall sensor; microSD; protected 12 V→5 V supply with Li-ion backup; PCB and waterproof enclosure	17,000–36,000
Monitor add-on	1.3” OLED or 2” TFT, buzzer, push button, ID EEPROM	1,500–3,000
Emission add-on	MQ-7 (CO), MQ-2 (HC), NDIR CO_2 sensor (0–20 % range), IR smoke-opacity cell, BME280, ADS1115, pump, probe, cooling tube, water trap	8,000–18,000
Cloud and data	Free-tier cloud (MQTT, database, auth, push) and prepaid data SIM	≈1,000/month
Total prototype (all modules)		**26,500–57,000**

Prices are indicative and will be confirmed against local suppliers.

5. METHODOLOGY AND TIMELINE (16 WEEKS)

Weeks	Work package	Deliverable
1–2	Requirements and survey	Target bikes, emission limits and test procedure, final parts list
3–5	Base hardware and firmware	Sensor drivers, 4G/GNSS link, MQTT telemetry, offline buffer
6–7	Event and crash detection	Rule engine, tilt filter, countdown, SMS fallback, data logging
8–9	Cloud and accounts	Broker, database, authentication, role-based access, QR/invite linking, push alerts
10–11	Mobile app	Single role-based app: live map, alerts, ride history, analytics
12–13	Plug-and-play add-ons	Expansion bus and auto-detection, Monitor and Emission modules
14–15	Testing and evaluation	Road trials, drop tests, emission comparison at a test centre
16	Documentation and demo	Final report, demonstration video, live demo

6. EVALUATION PLAN

- **Crash detection:** precision, recall and false alarms per riding hour (potholes, parked tip-overs, drops).
- **Telemetry:** GPS speed vs. reference speedometer; end-to-end alert latency (device → guardian).
- **Emission pre-check:** agreement of the pass/fail estimate and gas readings with official test results on several bikes.
- **System:** add-on detection time and reliability, power draw, offline-to-online data recovery.

7. RISKS, ETHICS AND MITIGATION

Risk	Mitigation
Crash false alarms	Multi-condition logic, cancel countdown, tuning with real ride data
Low-cost gas sensor accuracy	Temperature/humidity compensation, calibration against test-centre readings, presented as an indicator only
Hot, wet exhaust damaging sensors	Cooling tube, water trap and filter; sensors kept outside the exhaust stream
Vibration, weather, electrical noise	Protected supply, sealed enclosure and connectors, backup battery
Poor cellular coverage	On-device detection, microSD buffering, SMS fallback
Privacy of tracked riders	Informed consent, rider sees who can view data, owner controls guardian access

8. EXPECTED OUTCOMES

A working prototype that (i) tracks a motorcycle and alerts guardians of possible crashes and risky riding without relying on the rider’s phone, (ii) supports hot-pluggable add-on modules detected automatically, and (iii) gives owners a low-cost emission pre-check before the official test, together with measured accuracy results and a clear path to a custom PCB product.